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SDC Hackathon

Phase 1 : Preliminary Project Proposal

Hosted by Xiamen University Malaysia (XMUM) AI Club

Submission Deadline: 25 October 2026

Instructions: Complete every field below and submit as a single PDF document named "SDC2026_TeamName_Preliminary.pdf" before the deadline above. Answers must comply with the word limits stated for each open-ended question. Incomplete submissions may not be eligible for review.

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1. Team & Project Overview

Team Name	WINNIE THE POOH
Selected Domain	1
Selected SDG Track (1–17)	4 & 8
Project / Solution Name	WinHire
Name (Leader)	Lim San Yee
Email & Phone (Leader)	eazen32@gmail.com
University Affiliation(s)	University of Nottingham Malaysia, Heriot-Watt University Malaysia, Monash University Malaysia
Number of Team Members	5

2. Executive Summary & Problem Statement

2.1 The Problem Context

Describe the specific, real-world challenge your team is tackling. Who is affected by this problem, and why does it matter? Highlight the critical social, environmental, or economic issues involved.

[Maximum 250 words]

As artificial intelligence (AI) transforms the employment landscape, Malaysian university students and fresh graduates face increasing challenges in identifying suitable career opportunities, understanding evolving skill requirements, and preparing for future employment. Existing recruitment platforms often contain outdated job advertisements, irrelevant applications, and potentially fraudulent employers. Furthermore, fragmented recruitment processes require users to switch between multiple applications for communication, interview scheduling, and career planning, reducing efficiency and transparency.

To address these challenges, WinHire is proposed as an AI-powered, all-in-one recruitment and career development platform that connects job seekers with employers through a secure, efficient, and integrated digital ecosystem.

WinHire introduces a minimum 50% job-matching threshold to improve application relevance, automatic job-post expiration after six months, company verification targeted within three working days, and supporting-document uploads to strengthen credibility. The platform also integrates direct messaging, virtual interview rooms, interview scheduling, and calendar reminders to streamline recruitment.

Beyond job applications, WinHire provides labor-market trend insights, Sustainable Development Goal (SDG) information, and AI-generated personalized learning roadmaps to help users identify skill gaps and prepare for evolving industry demands.

Aligned with SDG 4 (Quality Education) and SDG 8 (Decent Work and Economic Growth), WinHire aims to enhance recruitment efficiency, promote trustworthy employment opportunities, and empower job seekers to make informed career decisions while developing future-ready skills.

2.2 The Proposed AI Solution

Provide a high-level overview of your proposed application. How does it apply artificial intelligence to solve the problem stated above? Briefly mention the tools or technical skills gained from the SDC Phase 1 workshops (e.g., GenAI APIs, database architecture, cloud hosting) that you intend to leverage.

[Maximum 300 words]

Our proposed application is an AI-assisted career exploration and reskilling platform for Malaysian final-year students and fresh graduates. It connects a learner's current skills, interests and weekly study budget to explainable career options and practical learning actions.

The matching engine combines TF-IDF lexical relevance, weighted skill coverage and dated job-market signals to rank a curated catalogue of roles. Each recommendation identifies existing strengths, missing skills and supporting sources. Public job listings and AI-software release feeds provide updatable evidence, while a separately selectable historical Malaysian pilot offers limited local context.

Users compare two career directions and explore moderate and fast AI-adoption scenarios. These scenarios express alternative assumptions rather than certain predictions. The Your Next Best Skill feature compares hypothetical fit improvements against prerequisite-inclusive study effort, helping learners select a manageable next action.

The platform converts that choice into an ordered learning plan with real resources, estimated hours, weekly allocations and an SDG-linked practical project. Saved plans preserve the original evidence, while portfolio stages distinguish self-reported progress, submitted project references and reviewer feedback. A guided assistant explains the recommendations and their limitations. An optional OpenAI adapter can classify questions when configured and explicitly consented to; it is disabled by default.

The implementation uses an HTML/CSS/JavaScript interface, Python Flask APIs and SQLite storage. We intend to apply relevant Phase 1 workshop learning in API integration, database architecture and deployment to strengthen the prototype. It currently runs locally, with public cloud deployment remaining a future step.

Career choices and final hiring decisions remain under human control; scores and occupational exposure indicators are not guarantees of employment or predictions of personal job loss.

3. Technical Architecture & Implementation Plan

3.1 Conceptual Tech Stack

List the technologies, frameworks, APIs, and databases you plan to use to build a functional, deployed prototype. Describe (or attach as an appendix) a simple system block diagram showing how the workshop elements connect into one architecture.

Did your project utilize any official SDC sponsor-provided GenAI API or Cloud Platform? If yes, specify which one and how it is used.

The prototype uses the following technology stack:

Component	Technology / Service	Purpose
Frontend	HTML, CSS and JavaScript	Responsive interfaces for career exploration, learning plans and recruitment workflows.
Backend	Python and Flask	API endpoints, validation, authentication, permissions and application logic.
Recommendation engine	Locally implemented TF-IDF, weighted skill matching and scenario-based scoring	Explainable role recommendations, skills-gap analysis and learning prioritisation.
Database	SQLite	Stores account records, applications, messages, interviews, saved plans and portfolio feedback.
Guest storage	Browser local storage	Retains guest learning progress separately from account data.
Evidence sources	Public job-listing JSON and Hugging Face Transformers Atom feeds	Supplies dated, updatable job and AI-development signals, with cached fallback.
Malaysian evidence	Bundled historical Malaysian job pilot	Provides explicitly limited local context, separate from international evidence.

Optional GenAI integration	OpenAI Chat Completions API	Question-to-intent classification when configured and consented to; disabled by default.
Version control	Git and GitHub	Maintains source code, development history and judge-accessible repository materials.
Testing	pytest and Chromium/Playwright	Checks backend behaviour and desktop/mobile interactions.
Deployment	Local Flask prototype; public hosting planned	A Python-compatible cloud host with persistent storage will be selected for public deployment.

Conceptual system architecture

User Interface — HTML / CSS / JavaScript

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Python Flask Backend API

├── Recommendation Engine

│ TF-IDF + skills + scenarios

├── SQLite Database

├── Public Evidence Adapters

│ Job/AI feeds + dated cache

└── Optional OpenAI Adapter

Disabled by default

Guest learning progress → Browser local storage

Current deployment: Local computer

Planned deployment: Public Python hosting with persistent storage

This architecture connects API integration, recommendation logic, database design and deployment within one application. Public hosting must preserve SQLite data and server-side secrets.

Official sponsor resources: No official SDC sponsor-provided GenAI API or cloud platform is used in the current prototype.

3.2 Design-Thinking & Deployment Strategy

Explain how your team applied design-thinking methodology to ensure this solution is technically viable, user-friendly, and ready for real deployment by Phase 2.

[Maximum 200 words]

Our design-thinking process began with team discussions and the Domain 1 brief to identify a specific need: Malaysian graduates deciding which skills to develop within limited time and resources. We defined the core journey as **profile → evidence → career comparison → next learning action**, keeping recruitment functions as supporting features.

During ideation and prototyping, we prioritised explainable scores, visible skills gaps, prerequisite-inclusive study estimates and meaningful SDG connections. Iterative interface changes introduced focused tabs, saved-plan continuity and a consistent Slate & Ice Blue theme. Responsive layouts, keyboard focus and clear loading/error states support usability.

Technical testing uses automated checks and desktop/mobile browser journeys with synthetic profiles. Planned anonymous task walkthroughs and independent adviser review will assess whether users understand recommendations and complete the core journey.

For Phase 2, we plan to deploy through a production-capable Python server with HTTPS, persistent database storage, server-side configuration and backups. Deployment checks will cover authentication, permissions, source failures, data persistence and judge access. Dated caches will support continuity during feed outages. The initial deployment will retain synthetic demonstration data and explicit limitations while further validation informs wider use.

4. Preliminary Environmental & SDG Impact Framework

To align with SDC's focus on meaningful global sustainability, establish your project's impact baseline before building the prototype. Use estimated data, published research, or public benchmarks where exact figures are unavailable. Provide the citation of any sources used.

The target, baseline, and KPI you provide here will be used by judges during shortlisting to assess ***Innovation and Creativity*** *as well as* ***Feasibility of build****. Vague or unsupported projections may weaken your case for advancing to Phase 2.*

Metric Category	**Your Response**
Target SDG Sub-Target(s) *Identify the exact UN SDG sub-target your project addresses* *e.g., Target 11.6 "Reduce the adverse per-capita environmental impact of cities, focusing on air quality and waste management."*	**Primary — SDG Target 4.4:** "By 2030, substantially increase the number of youth and adults who have relevant skills, including technical and vocational skills, for employment, decent jobs and entrepreneurship." The platform identifies skills gaps and provides personalised learning plans, practical projects and feedback opportunities. **Supporting — SDG Target 4.3:** Ensure equal access to affordable, quality technical, vocational and tertiary education. Guest access and free or low-cost learning resources support affordable career exploration. **Supporting — SDG Target 8.5:** Achieve full and productive employment and decent work for all, including young people. Explainable role comparisons, salary transparency and application tracking support informed career decisions. Employment outcomes remain to be evaluated.
Current Problem Baseline *What is the starting condition of the problem, expressed as a number?* *e.g., "The target infrastructure currently experiences 35% energy inefficiency due to unoptimized scheduling."*	**Published benchmark:** The International Labour Organization's 2025 study estimates that approximately **24% of global employment** is in occupations with some exposure to generative AI. This establishes the scale of potential changes to work and the relevance of reskilling. Exposure does **not** mean job loss, and this global figure is **not a Malaysian fresh-graduate estimate**. **Project baseline:** No completed graduate-user outcome evaluation has been established. Before pilot testing, we will measure participants' time to select a suitable role, ability to identify relevant skills gaps and ability to define a practical

Metric Category	Your Response
	next learning action. These measurements will provide the comparison baseline for evaluating the platform.
Core Measurable KPI *The specific unit you will track* *e.g., % reduction in power consumption, processing-latency improvement, or kg of resource waste avoided.*	**Percentage reduction in median time required to create a complete career action plan.** A complete plan must identify one suitable role, two relevant skills gaps and one learning action that fits the participant's stated time and budget. Measure task time in minutes when participants use conventional browsing and when they use the platform. Use comparable fictional profiles and the same source material, alternating the testing order to reduce practice effects. **Calculation:** [(baseline median time − platform median time) ÷ baseline median time] × 100. Report task completion rates alongside timing so faster but incomplete plans do not count as success.
Phase 2 Impact Target *Your realistic, percentage-based target by the final prototype presentation* *e.g., "20% improvement in computational resource use."*	**Aim for at least a 20% reduction in median career-planning time** by the final prototype presentation, evaluated through a small pilot with **10 volunteer students or recent graduates**. As a quality safeguard, at least **80% of participants — 8 out of 10 should produce a complete plan** meeting the predefined checklist. Record incomplete tasks, participant feedback and any disagreements about recommended skills. These are team-defined feasibility targets; actual results will be reported after testing, including whether either target was missed.

5. Eligibility & Readiness Checklist

Tick all boxes that apply before submitting.

- ☐ All team members have completed the mandatory GenAI, Database, and Cloud Deployment upskilling workshops.

- [/] The project concept addresses a verified real-world problem mapped directly to one or more UN SDGs.

- [/] The proposed solution targets a functional, deployed application rather than a theoretical business case study.

- [/] All fields in this template have been completed within the word limits stated.

6. IP & Originality Declaration

6.1 Pre-Existing / Prior-Owned Work

Does this submission include any code, design, research, or other work that any team member created or owned before the start of this hackathon?

☐ Yes ✅ No

If yes, briefly describe what was pre-existing and which team member(s) owned it:

6.2 Third-Party Code, Datasets, or Models

Does your project use any third-party code libraries, open-source frameworks, public/private datasets, or pre-trained AI models that your team did not create?

✅Yes ☐ No

If yes, list them below (open-source use is permitted provided it is declared and properly licensed/attributed):

Name	Type (code/dataset/model)	License
Python and standard library	Code — backend runtime	Python Software Foundation License; applicable component notices
Flask 3.1.2	Code — web framework	BSD-3-Clause
Werkzeug, Jinja2, MarkupSafe, itsdangerous and Click	Code — Flask dependencies	BSD-3-Clause; retain each package's notices
Blinker	Code — signalling dependency	MIT

tzdata	Code/data — timezone support	Apache-2.0; timezone-data source notices also apply
SQLite	Code — database engine	Public domain
pytest, iniconfig and pluggy	Code — automated testing	MIT
packaging	Code — testing dependency	Apache-2.0 OR BSD-2-Clause
Pygments	Code — testing/support dependency	BSD-2-Clause
CareerTrends Malaysian job-market pilot	Dataset — attributed extract of 16 historical Malaysian job observations	MIT
ILO–NASK 2025 occupational exposure research	Dataset — published occupational exposure values	CC BY 4.0; attribution and disclosure of adaptations required
OpenDOSM labour-force context	Dataset — source metadata and optional aggregate observations	Official data: CC BY 4.0; selected metadata: MIT. Numeric retrieval was unavailable in the inspected environment.
SimplifyJobs/New-Grad-Positions	Dataset, public international job-listing feed	A dataset reuse licence has not been verified; source terms require confirmation

Hugging Face Transformers release feed	Dataset — public software-release signals	Feed-content reuse terms require confirmation; the repository's code licence does not automatically cover all release content
Frey–Osborne, *The Future of Employment* (2013)	Dataset — cited historical appendix values	No explicit open reuse licence identified; factual values are cited and the full paper is not redistributed
Our World in Data energy data	Dataset — linked practical-project reference	OWID-created material: CC BY 4.0; underlying datasets retain source-specific terms

6.3 Employment-Related IP Conflicts

Is any team member currently bound by an employment, internship, or consulting agreement under which their employer could claim ownership of work they create during this hackathon?

☐ Yes ✅ No

If yes, please describe the nature of the agreement and which member it concerns:

7. Declaration

By submitting this proposal, the team leader confirms that the information provided is accurate to the best of the team's knowledge and that the project will be original work created during the SDC 2026 hackathon period.

XinRu Sanyee WinVien JingHeng Ophelia

25/10/2026